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ASSIGNMENTS

# Problem set 4

AI504 — Knowledge Representation

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# 1 Problem 1

Do exercise 2.3 from the textbook, reproduced herewith. Work with the set of nouns hawks, birds, turtles and the verb see. Let  $\Gamma$  be the set of sentences

- all hawks birds
- all (see all turtles) (see all (see all hawks))
- all (see all (see all hawks)) turtles
- all turtles (see all birds)

Give a formal proof (in either  $\mathcal{A}(\mathcal{RC})$  or  $\mathcal{A}(\mathcal{RC})'$ ) of each of the following sentences from  $\Gamma$ . (Attach more pages if space demands)

- (a) all (see all (see all hawks)) (see all hawks)
- (b) all (see all turtles) (see all (see all birds))
- (c) all (see all turtles) (see all (see all turtles))

## 1.1 Solution

For this problem nouns will be addressed as abbreviations for simplicity and spacing.

hawks = h, birds = b, turtles = t

a)

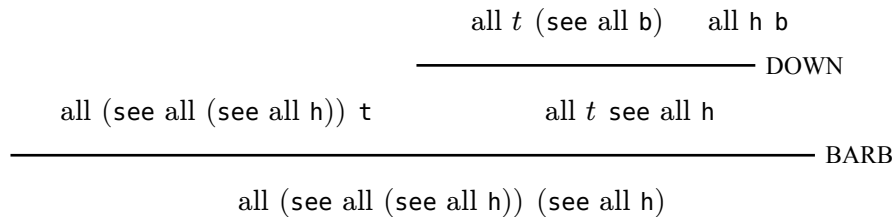


Figure 1: proof tree of (a)

b)

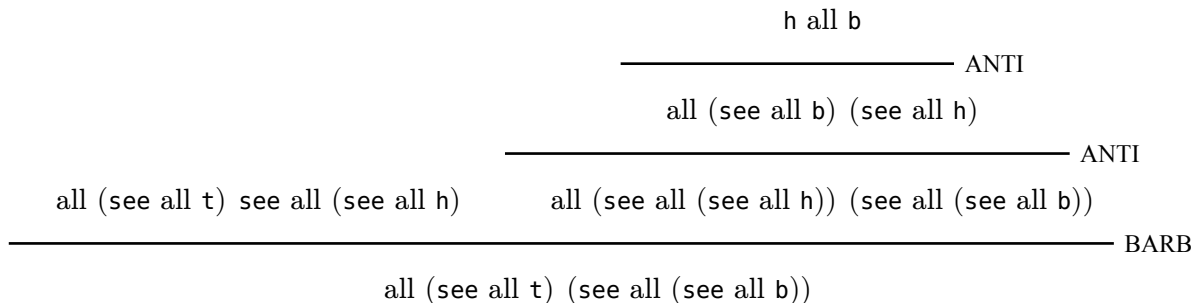


Figure 2: proof tree of (b)

c)

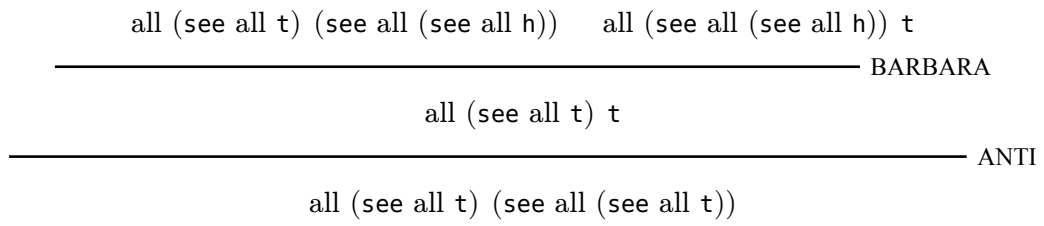


Figure 3: proof tree of (c)

## 2 Problem 2

This problem lets you practice *induction on terms*. Work with the set of verbs `see`, `love`, and one noun `cat`. We get terms like

`love all (see all (see all (love all (love all cat))))`

This term starts with one occurrence of `love`. The term `cat` and `see all (love all cat)` start with zero occurrences of `love`. And `love all (love all cat)` starts with two occurrences of `love`.

Let's take a model  $\mathcal{M}$  with  $M = \{1, 2\}$  and with

$$\llbracket \text{cat} \rrbracket = \{1, 2\}$$

$$\llbracket \text{see} \rrbracket = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

$$\llbracket \text{love} \rrbracket = \{(1, 2)\}$$

Prove that for any term  $t$ , the denotation  $\llbracket t \rrbracket \subseteq M$  is either  $M$  itself or  $\emptyset$ , depending on whether the number of occurrences of `love` that  $t$  starts with is either even or odd.

## 2.1 Solution

### Proof by induction

**Goal:** For any term  $t = (v \text{ all } t)$ ,  $\llbracket t \rrbracket \subseteq M$  is either  $M$  or  $\emptyset$  depending on whether the number of occurrences of  $\text{love}$  that  $t$  starts with is either even or odd.

#### Inductive hypothesis

Since  $\llbracket r \text{ all } t' \rrbracket = \{m \in M : \forall n \in \llbracket t' \rrbracket (m, n) \in \llbracket r \rrbracket\}$ , any term on the form  $t = \text{see all } t'$

Then since  $\llbracket \text{see all } t' \rrbracket = M$

Because of this any term can be written by definition as

$$t = \text{love}^k(t') \quad k \in \mathbb{N}$$

where  $t' \neq \text{love all } t$

#### Base case ( $k = 0$ )

Term  $t$  can either be a single noun or an expression

Case 1

Term  $t$  is single noun ( $\text{cat}$ ) with 0 (even) occurrences of  $\text{love}$

$$\llbracket \text{love}^0(\text{cat}) \rrbracket = \llbracket \text{cat} \rrbracket = \{1, 2\} = M.$$

Case 2

Term  $t$  is ( $\text{see all cat}$ ) with 0 (even) occurrences of  $\text{love}$

$$\llbracket \text{love}^0(\text{see all cat}) \rrbracket = \llbracket \text{see all cat} \rrbracket = \{1, 2\} = M$$

#### Inductive step ( $k + 1$ )

By **IH** lets only focus on leading  $\text{love}$ 's

For  $2k + 1$

|  $\llbracket t' \rrbracket = M$  then since  $\llbracket \text{love} \rrbracket = \{(1, 2)\}$ ,  $\forall n \in M$  will result in  $\emptyset$

For  $2k$

|  $\llbracket t' \rrbracket = \emptyset$  then the universal quantifier  $\forall n \in \emptyset$  makes  $\llbracket t \rrbracket = M$

Because of this,

$$\text{love}^{2k}(t') = M \text{ (even)} \quad \text{love}^{2k+1}(t') = \emptyset \text{ (odd)}$$

when  $t' \neq \text{love all } t$

### 3 Problem 3

Let  $\Gamma = \{\text{all } p \ q\}$ . For each of the following two sentences give *either* a proof (in either  $\mathcal{A}(\mathcal{RC})$  or  $\mathcal{A}(\mathcal{RC})'$ ) of that sentence from  $\Gamma$  *or* a counter-model (i.e., a model which satisfies all  $p \ q$ , but falsifies the sentence in question). If you're giving a counter-model, verify that your counter-model actually works, i.e., satisfies and falsifies the right sentences, and show your work.

(a)  $\text{all } (\text{hate all } (\text{love all } p)) \ (\text{hate all } (\text{love all } q))$

(b)  $\text{all } (\text{hate all } (\text{love all } q)) \ (\text{hate all } (\text{love all } p))$

#### 3.1 Solution

a) One can show that  $\Gamma \vdash \phi$  by using the (Anti) twice, while treating the sentence (a) as a leaf and building up the tree

$$\begin{array}{c}
 \text{all } p \ q \\
 \hline
 \text{all } (\text{love all } q) \ (\text{love all } p) \quad \text{ANTI} \\
 \hline
 \text{all } (\text{hate all } (\text{love all } p)) \ (\text{hate all } (\text{love all } q)) \quad \text{ANTI}
 \end{array}$$

Figure 4: proof tree

b) Creating a counter example:

$$\begin{aligned}
 M &= \{1, 2, 3, 4\} \\
 [[p]] &= \emptyset \\
 [[q]] &= \{1, 2\} \\
 [[\text{love}]] &= \{(1, 1), (2, 2)\} \\
 [[\text{hate}]] &= \{(1, 1), (2, 2), (3, 3), (4, 4)\}
 \end{aligned}$$

first we have to check that our model satisfies  $\phi$ :  $m \models \text{all } p \ q \checkmark$

Taking the right side:

$[[\text{love all } q]] = \{m \in \{1, 2, 3, 4\} : \forall n \in \{1, 2\}, (m, n) \in \{(1, 1), (2, 2)\}\}$ . This is  $\emptyset$  set since we don't have all the pairs mapping from  $[[p]]$  to  $[[\text{love}]]$ .

$[[\text{hate all } (\text{love all } q)]] = \{m \in \{1, 2, 3, 4\} : \forall n \in \emptyset : (m, n) \in \{(1, 1), (2, 2), (3, 3), (4, 4)\}\} = M$  since we are walking with a universal quantifier mapping over the empty set, which is always true.

Taking the left side:

$[[\text{love all } p]] = \{m \in \{1, 2, 3, 4\} : \forall n \in \emptyset, (m, n) \in \{(1, 1), (2, 2)\}\} = M$ .

$[[\text{all } (\text{hate all } (\text{love all } p))]] = \{m \in \{1, 2, 3, 4\} : \forall n \in \{1, 2, 3, 4\}, (m, n) \in \{(1, 1), (2, 2), (3, 3), (4, 4)\}\} = \emptyset$ .

$\text{all } (\text{hate all } (\text{love all } q)) \ (\text{hate all } (\text{love all } p)) = \{1, 2, 3, 4\} \not\subseteq \emptyset$ .

Therefore one can **conclude**:  $\mathcal{M} \not\models \text{all } (\text{hate all } (\text{love all } q)) \ (\text{hate all } (\text{love all } p))$ .

## 4 Problem 4

This problem continues the notion of duals from homework 2. We can define the *dual* of the sentence  $\text{all } s \text{ } t$  to be sentence  $\text{all } t \text{ } s$ . So for example, for the dual of the sentence  $\text{all } t \text{ } (\text{see all } b) \text{ } t$  (see  $\text{all } b$ ) is  $\text{all } (\text{see all } b) \text{ } t$ . Let  $\phi^\dagger$  denote the dual of  $\phi$ . Note that this *extends* the previous notion of dual to sentences in  $\mathcal{A}(\mathcal{RC})$

My question is: if  $\Gamma \vdash \phi$ , is it still true that  $\Gamma^\dagger \vdash \phi^\dagger$ ? You must either **yes** and include on proof by induction on proof trees (like before), or **no** and a counterexample. (A counterexample consists of a *specific* set of sentences  $\Gamma$  and a sentence  $\phi$  and a *specific* model  $\mathcal{M}$  such that  $\Gamma \vdash \phi$  and  $\mathcal{M} \models \Gamma^\dagger$  but  $\mathcal{M} \not\models \phi^\dagger$ .) Note that if your answer is **yes**, you can work either with  $\mathcal{A}(\mathcal{RC})$  or  $\mathcal{A}(\mathcal{RC})'$ , as both give equivalent notions of drivability  $\vdash$ . But you should be clear which one you're choosing.



## 4.1 Solution

### Proof by induction

**Goal:** For all proof trees  $T$ , where  $\Gamma \vdash \phi$ , then there exists a proof tree  $T^\dagger$ , such that  $\Gamma^\dagger \vdash \phi^\dagger$ .

#### Base case

If  $T$  is a single sentence (the whole tree is a leaf), then either  $\phi$  is an axiom such that  $\phi = \phi^\dagger$  or  $\phi \in \Gamma$  which means that  $\phi^\dagger \in \Gamma^\dagger$

Then since all leaves are  $\in \Gamma^\dagger$ ,  $\Gamma^\dagger \vdash \phi^\dagger$

#### Inductive hypothesis

For all proofs  $T$  where  $\Gamma \vdash \phi$ , there exists another proof  $T'$  from  $\Gamma^\dagger$  such that  $\Gamma^\dagger \vdash \phi^\dagger$

#### Inductive step

Suppose  $T$  is not a leaf, then it is either an instance of ANTI or BARBARA.

Case 1 ANTI:

If  $T$  is not a single leaf but an instance of ANTI, then it is a sentence proved by 1 subtree. This can be written up as follows:

$$\frac{\frac{\mathcal{T}}{\text{all } x \ y}}{\text{all } (r \ \text{all } y) \ (r \ \text{all } x)} \text{ANTI}$$

$\phi$

Where  $\phi^\dagger$  can be proved by using ANTI on the dual of the subtree

$$\frac{\frac{\mathcal{T}^\dagger}{\text{all } y \ x}}{\text{all } (r \ \text{all } x) \ (r \ \text{all } y)} \text{ANTI}$$

$\phi^\dagger$

By the basis of the inductive hypothesis applied to  $\mathcal{T}$ , it follows that  $\Gamma^\dagger \vdash \mathcal{T}^\dagger$ , and by using the ANTI rule, it follows that  $\Gamma^\dagger \vdash \phi^\dagger$ .

Case 2: BARBARA

If  $\mathcal{T}$  is not a single leaf, but an instance of BARBARA, then it is a sentence proved by 2 subtrees  $\mathcal{T}_0, \mathcal{T}_1$  which can be written as follows:

$\phi$  and  $\phi^\dagger$ , but  $\mathcal{T}$  is composed of two trees, left and right trees labeled  $\mathcal{T}_0$  and  $\mathcal{T}_1$ , respectively, and it's dual:

$$\frac{\frac{\mathcal{T}_0}{\text{all } (p \ \text{all } x)} \quad \frac{\mathcal{T}_1}{\text{all } (x \ \text{all } q)}}{\text{all } p \ q} \text{BARBARA}$$

$\phi$

The proof tree can then be restructured, so it satisfies the  $\phi^\dagger$ .

By switching the order of the dual of the subtrees  $\mathcal{T}_0^\dagger, \mathcal{T}_1^\dagger$ ,  $\phi^\dagger$  can also be proved by an instance of BARBARA:

|    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | <div style="border-left: 1px solid black; padding-left: 10px; vertical-align: top;"> <div style="border-left: 1px solid black; padding-left: 10px; vertical-align: top;"> <math display="block">\frac{\text{all } \overbrace{(q \text{ all } x)}^{\mathcal{T}_1^\dagger} \overbrace{(x \text{ all } p)}^{\mathcal{T}_0^\dagger}}{\underbrace{\text{all } q \ p}_{\phi^\dagger}} \text{ BARBARA}</math> </div> </div> |
| 21 | <div style="border-left: 1px solid black; padding-left: 10px; vertical-align: top; padding-top: 10px;"> <p>Therefore, all proofs <math>T</math> from <math>\mathcal{A}(RC)'</math> where <math>\Gamma \vdash \phi</math>, proves the existence of <math>\Gamma^\dagger \vdash \phi^\dagger</math>.</p> </div>                                                                                                        |